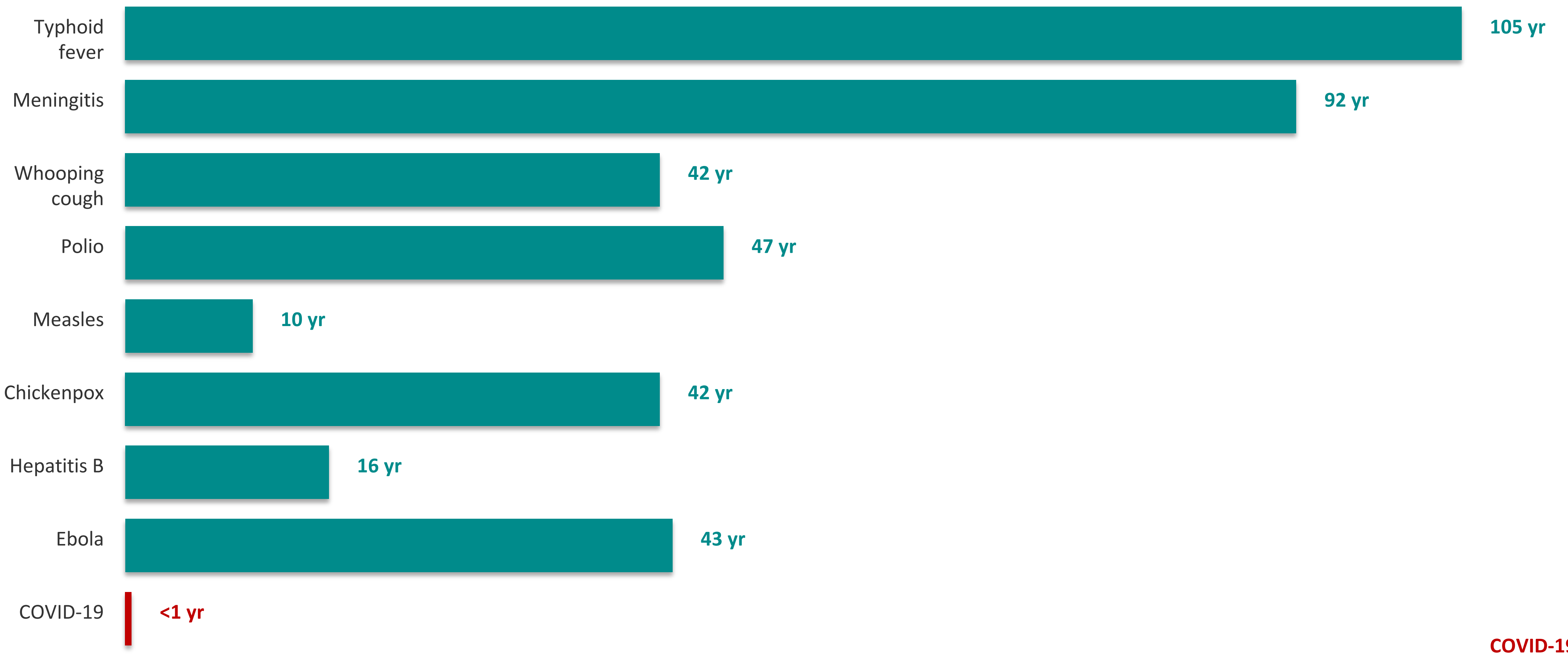
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of vaccine impact across two centuries (1796–2020)

Source: Our World in Data | Audience: Public Health Professionals & Policy Analysts

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



COVID-19: Record-breaking <1 year via mRNA technology

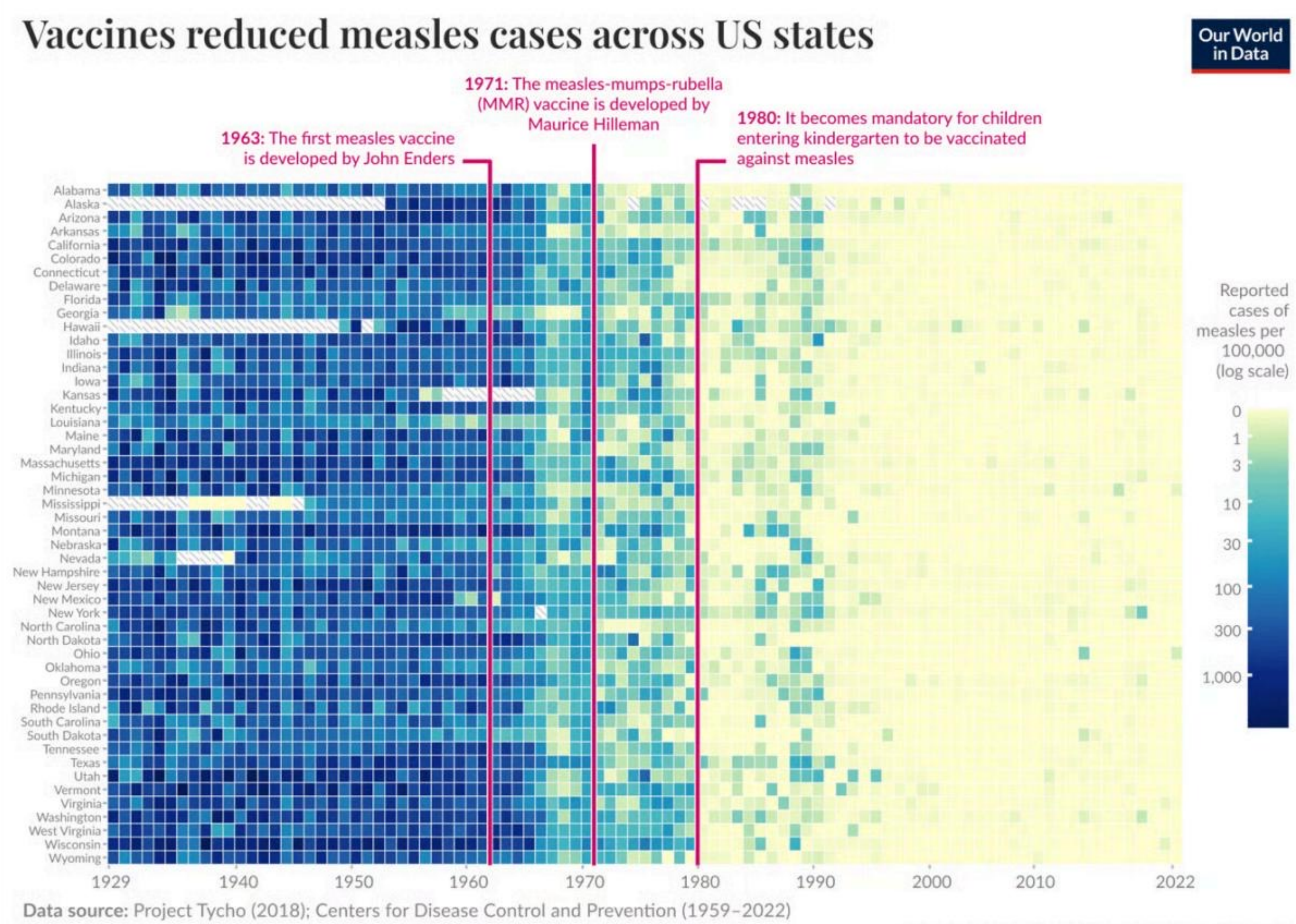
Source: Our World in Data

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases (per million/yr)	Post-Vaccine Cases (per million/yr)	Case Reduction	Pre-Vaccine Deaths (per million/yr)	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Source: Roush & Murphy (2007), Our World in Data

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



1963 First measles vaccine developed by John Enders

1971 MMR (measles-mumps-rubella) vaccine developed by Maurice Hilleman

1980 Vaccination became mandatory for children entering kindergarten

Before 1963: nearly every state had 100–1,000+ cases per 100,000. After 1980 mandate: cases dropped to near zero nationwide.

Smallpox: The Only Human Disease Eradicated Through Vaccination

1796 Edward Jenner creates the first vaccine using cowpox

1959 WHO passes resolution to eradicate smallpox globally

1967 WHO launches intensified eradication campaign

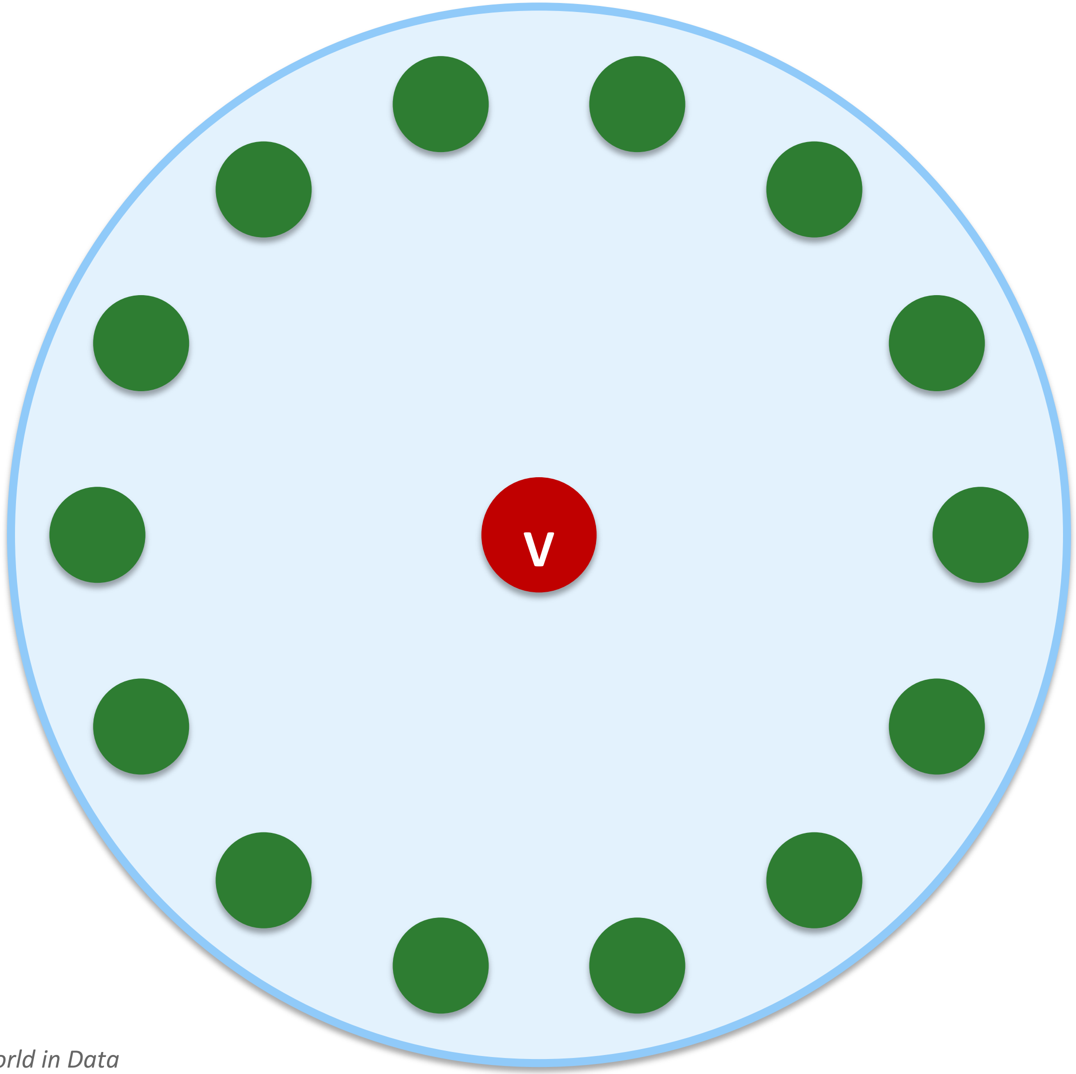
1977 Last natural case occurs in Somalia

1980 Smallpox declared eradicated worldwide — the only human disease fully eliminated through vaccination

Why smallpox was uniquely suited for eradication: infected only humans, symptoms were easily distinguishable, and the vaccine was cheap and highly effective.

Source: Our World in Data

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



How It Works

When enough people are vaccinated, pathogens cannot spread effectively through the population.

This creates a protective barrier around those who cannot be vaccinated:

- Newborns too young for certain vaccines
- Immunocompromised individuals
- Cancer patients on chemotherapy
- People with severe allergies to vaccine components

Source: Our World in Data

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Diseases with Incomplete Vaccine Impact

Pneumococcal disease

40.5% case reduction • 31.3% death reduction

Lowest impact among tracked diseases

Acute hepatitis B

83.9% case reduction • 83.6% death reduction

Significant burden remains

Hepatitis A

89% case reduction • 88.7% death reduction

Still requires targeted intervention

Systemic Barriers to Full Coverage

Access Gaps

Not everyone has access to vaccines, especially in low-income regions and remote areas.

Supply Shortages

Manufacturing and distribution bottlenecks put children at risk of missing critical doses.

Trust & Uptake

Poor scientific understanding and misinformation can reduce public confidence in vaccination.

Source: Our World in Data

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

1. **100% Elimination Achieved**

   Diphtheria, smallpox, and both forms of polio have been completely eliminated in the US through vaccination.

2. **Timelines Collapsed**

   Vaccine development accelerated from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.

3. **State-Level Measles Success**

   Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.

4. **Smallpox: A Global First**

   Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global WHO campaign.

5. **Gaps Still Persist**

Call to Action: Strengthen global health systems to improve vaccine access, close coverage gaps, and rebuild public trust in vaccination.

Source: Our World in Data